DATA LAKE ARCHITECTURE — Medical Data Processing Company

Open Source Stack | On-Premise Hadoop Ecosystem | HIPAA Compliant | Fault-Tolerant | Horizontally Scalable

DATA SOURCES

8,000 Medical Facilities

UC · ER · ICU · Hosp · NH

Data Formats

- **FTP / SFTP** — File Server
- **REST APIs** — API Agent
- **CSV**
- **XML**
- **TXT**

Data Volume

- 77,000 zip files/day
- 15M data files/day
- 3,500 zip files/hr
- 15-20% YoY growth

Legacy SQL Server
(Migration Source)
Apache Sqoop → HDFS

End Users

Analyst · Engineer · Customer

① INGESTION LAYER

Apache NiFi

Visual Dataflow Mgmt
- FTP/SFTP polling
- Gzip decompression
- Backpressure control
- File validation

Replaces: File Extract Agent

Apache Kafka

Message Queue
- Stream ingestion
- Topic partitioning
- Message persistence
- Replay on failure

3,500 zip files/hr

Debezium

Kafka Connect

Change Data Capture (CDC)
- SQL Server logs
- Row-level changes
- UPSERT support
- Zero app changes

Replaces: SSIS CDC

CDC events

Apache Sqoop

Bulk DB Import
- SQL Server → HDFS
- Parallel import
- Incremental load
- Initial migration

Replaces: SSIS Bulk

Apache Flume

Log & File Collection
- Agent-based collect
- Channel buffering
- Reliable delivery
- Facility endpoints

Replaces: API Agent

Ingest to HDFS

⑤ CROSS-CUTTING CONCERNS

SECURITY & GOVERNANCE

Apache Ranger + Knox

- Apache Ranger RBAC — Row & Column-level security
- At-Rest Encryption — HDFS TDE — AES-256
- In-Transit Encryption — TLS/SSL all data movement
- Kerberos Auth — All services & users
- Apache Knox Gateway — Perimeter API gateway
- HIPAA Compliance — PHI classification via Atlas; Audit trail in Raw Zone

HA & DISASTER RECOVERY

Replaces: Manual SQL Backups

- HDFS Replication Factor: 3 — Auto fault tolerance, 3 data nodes
- NameNode HA (ZooKeeper) — Active/Standby auto failover
- Apache DistCp Replication — Cross-cluster nightly backup
- ZooKeeper Coordination — Leader election, cluster state

RTO Improvement
Before: Hours-Days (SQL restore)
After: Sub-minute (auto failover)

Node Failure Handling
HDFS: Automatic, zero downtime
SQL: Manual, full outage

MONITORING & OPERATIONS

Ambari · Prometheus · Grafana

- Apache Ambari / Cloudera Mgr — Cluster health & config
- Prometheus + Grafana — Metrics, alerts, dashboards
- ELK Stack (Elasticsearch) — Centralized log management
- Apache Airflow UI — Pipeline monitoring & alerts

Hadoop Cluster
Cloudera HDP (On-Premise)
YARN Resource Management
Horizontal scale-out nodes
Open source — no vendor lock
Scales with 15-20% YoY growth

② STORAGE LAYER — HDFS (Hadoop Distributed File System | Replication Factor: 3)

RAW ZONE

Landing Area | Immutable
Format: Parquet / ORC
- Original compressed files
- Schema-on-read
- Never deleted (audit trail)
- HIPAA compliant storage

Replaces: SQL Stage DB

STAGING ZONE

Cleaned & Validated
Apache Hudi (UPSERT)
- Dedup & cleansing
- Schema enforcement
- CDC UPSERT support
- Time-travel queries

Replaces: SQL Master DB

CURATED ZONE

Gold | Analytics-Ready
Parquet — Partitioned
- Aggregated tables
- Business-ready joins
- Patient metrics
- Compliance datasets

5+ year retention

ARCHIVE ZONE

Cold Storage | HIPAA 7yr+

🔥 HOT 0-6mo · ❄ COLD 6mo+

SSD → HDD → JBOD
Auto tiering policy
40-60% cost reduction
10+ year retention

METADATA STORE

Apache Atlas
- Data catalog & lineage
- Schema registry
- Business glossary
- PII/PHI classification
- Hive Metastore

Metadata-Driven Design

Process & Transform

③ PROCESSING LAYER

BATCH PROCESSING

Apache Spark (PySpark)
- Metadata-driven ETL jobs
- Replaces 70+ SSIS scripts
- Data cleansing & dedup
- Schema validation
- YARN resource mgmt

Apache Airflow (Orchestration)
Replaces nightly SSIS ETL
→ Runs on YARN cluster

Replaces: SSIS Packages

STREAM PROCESSING

Kafka Streams / Flink
- Near-real-time ingestion
- Live dashboard feeds
- Event-driven triggers
- Spark micro-batch
- Exactly-once semantics

Flink: complex event proc.
Sub-minute latency
→ Replaces nightly-only batch

New capability

AD-HOC & CDC

Hive · Impala · Presto
- Apache Hive SQL (batch)
- Impala: sub-second query
- Presto: federated SQL
- Apache Hudi CDC+UPSERT
- Time-travel queries

Cross-source SQL joins
No data movement needed
→ Replaces Stored Procs

Replaces: SQL stored procedures

ML & ANALYTICS (New Capability)

Spark MLlib
- Distributed ML on HDFS
- Patient risk scoring
- Anomaly detection
- Readmission prediction

No data movement needed

TensorFlow / MLflow
- Deep learning models
- MLflow model registry
- Jupyter / Zeppelin
- Experiment tracking

Model serving via REST

→ Enables CTO's ML roadmap — no data movement, no separate export
→ Directly addresses data silo problem: one lake, many engines

New capability: not possible with SQL Server

④ SERVING

SQL Reports

Hive / Impala
Daily reports
Weekly reports
Compliance
Replaces:
File exports

Dashboards

Superset
Grafana
PowerBI
Tableau
Real-time
Near-RT feeds

Web Portal REST APIs

FastAPI layer
Customer self-service portal
Bed availability
Load balanced

ML Model Serving

MLflow REST
Predictions
Risk scores
New capability

Ad-hoc Analytics

Presto/Trino
Jupyter NB
Federated SQL
Eliminates silos

Ingestion Layer · Storage Layer · Processing Layer · Serving Layer · Security / HA / Monitoring · Data Flow · Replaces legacy component

Stack: Apache NiFi · Kafka · Sqoop · Debezium · Flume · Hadoop HDFS · Hudi · Atlas
Spark · Flink · Hive · Impala · Presto · Airflow · Ranger · Knox · ZooKeeper · Superset